

MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 9: Conservative Schemes, Semi-Discretization, and Strong Stability Preserving (SSP) Methods

Instructor: Biswarup Biswas

1 Introduction

In Lectures 7 and 8, we studied fully discrete finite difference schemes for linear advection. When dealing with nonlinear hyperbolic conservation laws:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \quad (1)$$

solutions develop discontinuities (shock waves) regardless of how smooth the initial data begins.

Computing discontinuous solutions exposes two profound mathematical and computational challenges:

1. **Shock Speed Failure:** If a scheme is written in non-conservative form (e.g., using $u_t + uu_x = 0$ instead of $u_t + (u^2/2)_x = 0$), it converges to a discontinuous function that travels at the *wrong* shock speed, violating the Rankine-Hugoniot jump condition!
2. **Decoupling Space and Time:** In fully discrete schemes (like Lax-Wendroff), space and time are inextricably intertwined. To build modern high-order methods, we decouple spatial and temporal discretizations via the **Method of Lines (semi-discretization)**.
3. **Preserving Nonlinear Stability in Time:** Higher-order time integrators (standard Runge-Kutta) can introduce non-physical oscillations even if the spatial discretization is oscillation-free. This requires **Strong Stability Preserving (SSP)** time-stepping.

2 Conservative Schemes and the Lax-Wendroff Theorem

2.1 Conservative Form of Difference Schemes

A numerical scheme for the conservation law (1) is said to be in **conservative form** if it can be written as:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{F_{j+1/2}^n - F_{j-1/2}^n}{\Delta x} = 0 \quad (2)$$

where $F_{j+1/2}^n = \mathcal{F}(u_{j-p}^n, \dots, u_{j+q}^n)$ is the **numerical flux function** approximating the physical flux $f(u)$ at the cell interface $x_{j+1/2} = x_j + \Delta x/2$.

2.1.1 Properties of the Numerical Flux

For consistency with the differential equation, the numerical flux $\mathcal{F}$ must satisfy:

- (a) **Consistency:** When all arguments are identical, the numerical flux reduces to the physical flux:

$$\mathcal{F}(u, u, \dots, u) = f(u) \quad (3)$$

- (b) **Lipschitz Continuity:** $\mathcal{F}$ must be Lipschitz continuous in all its arguments.

- (c) **Telescoping Sum (Global Conservation):** Summing equation (2) over all spatial grid cells $j = 1, \dots, N$:

$$\sum_{j=1}^N u_j^{n+1} \Delta x = \sum_{j=1}^N u_j^n \Delta x - \Delta t (F_{N+1/2}^n - F_{1/2}^n) \quad (4)$$

All interior fluxes cancel telescopically! The total discrete mass $\sum_j u_j \Delta x$ changes solely due to the boundary fluxes entering and exiting the domain.

2.2 The Danger of Non-Conservative Schemes

By the chain rule, smooth solutions satisfy $f(u)_x = f'(u)u_x$. For inviscid Burgers' equation ($f(u) = u^2/2$):

$$\text{Conservative: } \frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{u^2}{2} \right) = 0 \quad \Longleftrightarrow \quad \text{Non-Conservative: } \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \quad (5)$$

These two forms are mathematically equivalent for smooth solutions, but **fundamentally distinct once a shock forms!** At a discontinuity, the derivative u_x is undefined in the classical sense. Discretizing the non-conservative form $u_j \frac{u_j - u_{j-1}}{\Delta x}$ lacks the telescoping property and creates artificial numerical mass sources at the shock front.

The Lax-Wendroff Theorem (1960)

Let u_j^n be the solution of a consistent, conservative numerical scheme (2) with grid parameters $\Delta t, \Delta x \rightarrow 0$. If u_j^n converges boundedly almost everywhere to a limit function $u(x, t)$, then:

$u(x, t)$ is a **weak solution** of the conservation law $u_t + f(u)_x = 0$.

In particular, any shock wave in the numerical solution travels with the **exact physical shock speed** given by the Rankine-Hugoniot jump condition:

$$s = \frac{f(u_L) - f(u_R)}{u_L - u_R}$$

Consequence: If a scheme is *not* conservative, even if it appears stable and converges as $\Delta x \rightarrow 0$, it can converge to a non-physical function with an entirely incorrect shock speed (demonstrated in Figure 1 Right).

3 Semi-Discretization: The Method of Lines (MOL)

Instead of discretizing space and time simultaneously, the **Method of Lines** discretizes the spatial derivative first, converting the PDE into a large coupled system of ordinary differential equations (ODEs).

3.1 Formulating the Semi-Discrete System

Let $u_j(t) \approx u(x_j, t)$ be the continuous-in-time numerical solution at node x_j . Discretizing the spatial flux derivative yields:

$$\frac{du_j}{dt} = \mathcal{L}_j(\mathbf{u}) = - \frac{F_{j+1/2}(t) - F_{j-1/2}(t)}{\Delta x} \quad (6)$$

In vector form for the entire grid $\mathbf{u}(t) = [u_1(t), \dots, u_N(t)]^T$:

$$\frac{d\mathbf{u}}{dt} = \mathbf{L}(\mathbf{u}) \quad (7)$$

3.2 Fourier Spectrum of Hyperbolic Semi-Discretizations

Consider the linear advection equation $u_t + au_x = 0$ ($a > 0$).

1. Central Differencing: $\frac{du_j}{dt} = -a \frac{u_{j+1} - u_{j-1}}{2\Delta x}$. Substituting $u_j(t) = \hat{u}(t)e^{ij\xi}$:

$$\frac{d\hat{u}}{dt} = \lambda(\xi) \hat{u}, \quad \lambda(\xi) = -i \frac{a}{\Delta x} \sin \xi \quad (8)$$

The spatial eigenvalues $\lambda(\xi)$ are **purely imaginary**, lying entirely on the imaginary axis $[\pm ia/\Delta x]!$

2. Upwind Differencing: $\frac{du_j}{dt} = -a \frac{u_j - u_{j-1}}{\Delta x}$:

$$\lambda(\xi) = -\frac{a}{\Delta x}(1 - e^{-i\xi}) = -\frac{a}{\Delta x}(1 - \cos \xi) - i \frac{a}{\Delta x} \sin \xi \quad (9)$$

Here $\text{Re}(\lambda) \leq 0$, representing negative real eigenvalues (spatial damping) shifted into the left half-plane.

4 Strong Stability Preserving (SSP) Runge-Kutta Methods

4.1 Why Standard Forward Euler Fails for Hyperbolic Central Differencing

If we integrate $\frac{d\hat{u}}{dt} = \lambda \hat{u}$ with Forward Euler: $\hat{u}^{n+1} = (1 + \lambda \Delta t) \hat{u}^n$. The stability region of Forward Euler is the disc $|1 + z| \leq 1$ in the complex plane $z = \lambda \Delta t$. Notice that the circle $|1 + z| \leq 1$ intersects the imaginary axis **only at the origin** $z = 0$. Because the central difference spectrum is purely imaginary ($z = -i\nu \sin \xi$), it lies entirely outside the Forward Euler stability domain for every $\nu > 0$. This gives an alternative, elegant proof of why FTCS is unconditionally unstable!

4.2 The Concept of SSP (TVD) Time-Stepping

Suppose a spatial semi-discretization $\frac{d\mathbf{u}}{dt} = \mathbf{L}(\mathbf{u})$ is stable (e.g., non-oscillatory / Total Variation Diminishing) under Forward Euler:

$$\|\mathbf{u} + \Delta t \mathbf{L}(\mathbf{u})\| \leq \|\mathbf{u}\| \quad \text{for } \Delta t \leq \Delta t_{\text{FE}} \quad (10)$$

A high-order multi-stage Runge-Kutta method is called **Strong Stability Preserving (SSP)** (originally called TVD Runge-Kutta by Chi-Wang Shu and Stanley Osher in 1988) if it can be written as a convex combination of forward Euler steps, ensuring that the nonlinear stability property of the spatial operator is preserved in time.

4.3 Popular SSP Runge-Kutta Schemes

4.3.1 Optimal Second-Order SSP Runge-Kutta: SSP-RK2 (Heun's Method)

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t \mathbf{L}(\mathbf{u}^n) \quad (11)$$

$$\mathbf{u}^{n+1} = \frac{1}{2} \mathbf{u}^n + \frac{1}{2} (\mathbf{u}^{(1)} + \Delta t \mathbf{L}(\mathbf{u}^{(1)})) \quad (12)$$

SSP Coefficient: $c_{\text{eff}} = 1$. The method is stable whenever $\Delta t \leq \Delta t_{\text{FE}}$.

4.3.2 Optimal Third-Order SSP Runge-Kutta: SSP-RK3 (Shu-Osher Scheme)

The workhorse time-stepper for modern high-resolution hyperbolic CFD (ENO/WENO):

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t \mathbf{L}(\mathbf{u}^n) \quad (13)$$

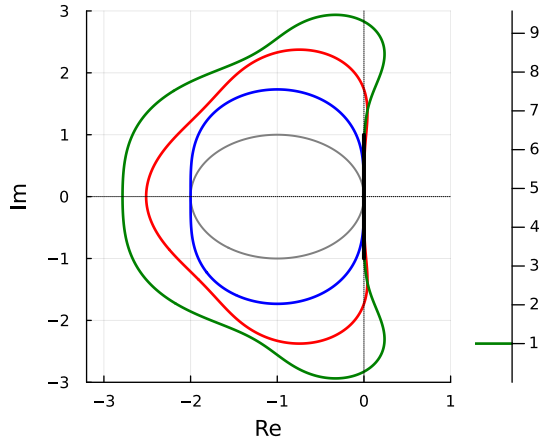
$$\mathbf{u}^{(2)} = \frac{3}{4} \mathbf{u}^n + \frac{1}{4} \mathbf{u}^{(1)} + \frac{1}{4} \Delta t \mathbf{L}(\mathbf{u}^{(1)}) \quad (14)$$

$$\mathbf{u}^{n+1} = \frac{1}{3} \mathbf{u}^n + \frac{2}{3} \mathbf{u}^{(2)} + \frac{2}{3} \Delta t \mathbf{L}(\mathbf{u}^{(2)}) \quad (15)$$

Properties:

- Third-order accurate in time ($\mathcal{O}(\Delta t^3)$).
- Convex combination of Forward Euler stages $\implies$ guaranteed monotonicity preservation.
- Its linear stability region encompasses a substantial segment of the imaginary axis $[-1.73i, 1.73i]$, allowing stable integration of central and high-order compact spatial stencils!

Stability Regions of RK / SSP-RK Methods



Lax-Wendroff Theorem: Shock Speed Failure

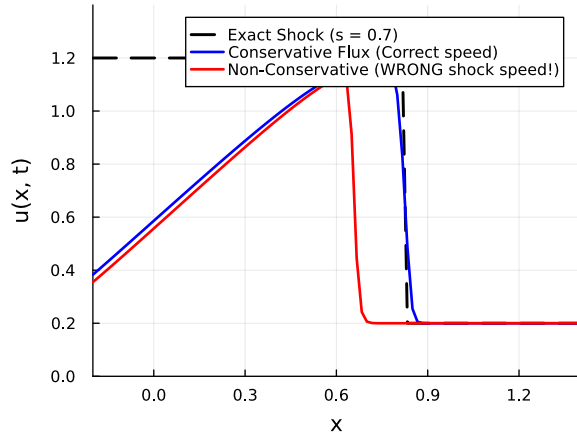


Figure 1: Left: Stability regions in the complex plane $\lambda\Delta t$. Forward Euler does not enclose the imaginary axis; SSP-RK3 and RK4 cleanly cover the hyperbolic advection spectrum. Right: Simulation of a shock wave for Burgers' equation ($u_L = 1.2, u_R = 0.2$). The conservative scheme moves at the exact theoretical shock speed $s = 0.7$, while the non-conservative formulation violates the Rankine-Hugoniot condition and computes an incorrect shock position. Generated with Julia.

5 Summary: The Trinity of Modern Hyperbolic Numerics

Concept	Mathematical Form	CFD Significance
Conservative Flux	$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x}(F_{j+1/2} - F_{j-1/2})$	Guarantees correct physical shock speed (Lax-Wendroff Theorem)
Method of Lines	$\frac{du_j}{dt} = -\frac{1}{\Delta x}(F_{j+1/2} - F_{j-1/2})$	Decouples high-order spatial differencing from temporal integration
SSP-RK Methods	Convex combinations of Forward Euler	Preserves spatial monotonicity and suppresses spurious wiggles

6 Exercises

Exercises for Lecture 9

E1. Analytical Verification of the Rankine-Hugoniot Shock Speed: Consider the Riemann problem for Burgers' equation with $u_L = 1.2$ and $u_R = 0.2$.

- (a) Using the Rankine-Hugoniot jump condition $s = \frac{f(u_L) - f(u_R)}{u_L - u_R}$, confirm that the exact shock speed is $s = 0.7$.
- (b) For the non-conservative discretization $u_t + uu_x = 0$, show why integrating across a jump $[u_L, u_R]$ fails to satisfy conservation of $\int u \, dx$.

E2. SSP-RK3 Imaginary Axis Intersection: For a linear single-harmonic ODE $\frac{du}{dt} = i\omega u$, show that the amplification polynomial for the 3rd-order Runge-Kutta scheme is:

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6}, \quad z = i\omega\Delta t$$

Evaluate $|R(iy)|^2$ for $y = \omega\Delta t \in \mathbb{R}$, and find the maximum value of y such that $|R(iy)| \leq 1$. (Confirm that the imaginary stability limit is $y \leq \sqrt{3} \approx 1.732$).

E3. Rusanov (Local Lax-Friedrichs) Numerical Flux: The Rusanov numerical flux for a scalar conservation law is defined by:

$$F_{j+1/2} = \frac{1}{2} (f(u_j) + f(u_{j+1})) - \frac{s_{\max}}{2} (u_{j+1} - u_j), \quad s_{\max} = \max(|f'(u_j)|, |f'(u_{j+1})|)$$

Prove that this flux is consistent with the physical flux $f(u)$.

References

- [1] John D. Anderson. *Computational Fluid Dynamics: The Basics with Applications*. McGraw-Hill, New York, NY, 1995.
- [2] Randall J. LeVeque. *Finite Volume Methods for Hyperbolic Problems*. Cambridge University Press, Cambridge, UK, 2002.